

## EXAM 7 – SPRING 2014

3. (1.25 points)

Given the following data for a Cape Cod reserve analysis:

### Actual Incremental Reported Losses (\$000)

| Accident<br><u>Year</u> | 12<br><u>Months</u> | 24<br><u>Months</u> | 36<br><u>Months</u> |
|-------------------------|---------------------|---------------------|---------------------|
| 2010                    | 100                 | 255                 | 180                 |
| 2011                    | 120                 | 280                 |                     |
| 2012                    | 120                 |                     |                     |

### Expected Incremental Reported Losses (\$000)

| Accident<br><u>Year</u> | 12<br><u>Months</u> | 24<br><u>Months</u> | 36<br><u>Months</u> |
|-------------------------|---------------------|---------------------|---------------------|
| 2010                    | 80                  | 300                 | 200                 |
| 2011                    | 80                  | 320                 |                     |
| 2012                    | 100                 |                     |                     |

The parameters of the loglogistic growth curve ( $\omega$  and  $\theta$ ) and the expected loss ratio (ELR) were previously estimated, resulting in a total estimated reserve of \$1,500,000. The parameter standard deviation of the total estimated reserve is \$350,000.

Calculate the standard deviation of the reserve due to parameter and process variance combined.

CONTINUED ON NEXT PAGE

## SAMPLE ANSWERS AND EXAMINER'S REPORT

| QUESTION 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                    |           |           |           |           |    |   |      |   |    |    |   |  |    |   |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|----|---|------|---|----|----|---|--|----|---|--|--|
| TOTAL POINT VALUE: 1.25                                                                                                                                                                                                                                                                                                                                                                                                                                                           | LEARNING OBJECTIVE: A3: Calculate the moments and percentiles of unpaid claim distributions implied by the models. |           |           |           |           |    |   |      |   |    |    |   |  |    |   |  |  |
| SAMPLE ANSWERS                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                    |           |           |           |           |    |   |      |   |    |    |   |  |    |   |  |  |
| Sample Answer 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                    |           |           |           |           |    |   |      |   |    |    |   |  |    |   |  |  |
| $\sigma^2 = \frac{1}{n - p} \sum \frac{(\text{Actual} - \text{Expected})^2}{\text{Expected}}$ <p>n = Number of points=6<br/>p=# parameters=2+1=3</p> $\sigma^2 = \frac{1}{6 - 3} \left( \frac{(100 - 80)^2}{80} + \dots + \frac{(180 - 200)^2}{200} \right)$ $= \frac{1}{3} * 42.75 = 14.25$ $14.25 * 1000 = 14,250$ $\text{Process Var} = \sigma^2 * \text{Reserve} = 14,250 * 1,500,000=2.1375 \times 10^{10}$ $\text{Total SD} = \sqrt{2.1375 * 10^{10} + 350000^2} = 379,309$ |                                                                                                                    |           |           |           |           |    |   |      |   |    |    |   |  |    |   |  |  |
| Sample Answer 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                    |           |           |           |           |    |   |      |   |    |    |   |  |    |   |  |  |
| R= 1,500,000<br>Parameter variance = (350,000) <sup>2</sup>                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                    |           |           |           |           |    |   |      |   |    |    |   |  |    |   |  |  |
| $\sigma^2 = \frac{1}{n - p} \sum \frac{(c - \mu)^2}{\mu}$ <p>n =6<br/>p=3</p> <table><tr><td><u>AY</u></td><td><u>12</u></td><td><u>24</u></td><td><u>36</u></td></tr><tr><td>10</td><td>5</td><td>6.75</td><td>2</td></tr><tr><td>11</td><td>20</td><td>5</td><td></td></tr><tr><td>12</td><td>4</td><td></td><td></td></tr></table> $\sigma^2 = \frac{1}{3} * 42.75 = 14.25$ $14.25 * 1000 = 14,250$ $\text{st dev} = \sqrt{1500000(14,250) + 350,000^2} = 379,309$             |                                                                                                                    | <u>AY</u> | <u>12</u> | <u>24</u> | <u>36</u> | 10 | 5 | 6.75 | 2 | 11 | 20 | 5 |  | 12 | 4 |  |  |
| <u>AY</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <u>12</u>                                                                                                          | <u>24</u> | <u>36</u> |           |           |    |   |      |   |    |    |   |  |    |   |  |  |
| 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5                                                                                                                  | 6.75      | 2         |           |           |    |   |      |   |    |    |   |  |    |   |  |  |
| 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 20                                                                                                                 | 5         |           |           |           |    |   |      |   |    |    |   |  |    |   |  |  |
| 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4                                                                                                                  |           |           |           |           |    |   |      |   |    |    |   |  |    |   |  |  |
| EXAMINER'S REPORT                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                    |           |           |           |           |    |   |      |   |    |    |   |  |    |   |  |  |
| Candidates generally performed well with the most common deduction for not reflecting the 1000's in the triangles. Some candidates used erroneous formulas for σ <sup>2</sup> or did not calculate the inputs correctly.                                                                                                                                                                                                                                                          |                                                                                                                    |           |           |           |           |    |   |      |   |    |    |   |  |    |   |  |  |